

INFO8006: Project 2 - Report

September 30, 2026

1 Bayes filter

- a. Let X_t be a ghost's cell, p_t Pacman's position, and $d(x, p) = |x_1 - p_1| + |x_2 - p_2|$. For requested variance v , put $n = \lfloor 4v \rfloor$ and $K_t \sim \text{Binomial}(n, 1/2)$. The measurement is $E_t = d(X_t, p_t) + K_t - n/2$. Therefore, with $k = e - d(x, p_t) + n/2$,

$$O(e \mid x, p_t) = \begin{cases} \binom{n}{k} 2^{-n}, & k \in \{0, \dots, n\}, \\ 0, & \text{otherwise.} \end{cases}$$

The sensor is unbiased and its actual variance is $n/4$, which need not equal the requested v . Measurements can be negative, and are half-integers when n is odd. No clipping or rounding is appropriate. Wall cells have zero prior mass and zero likelihood in the grid model.

- b. Let $N(x)$ be the free cardinal neighbors of x , including the reverse direction. Define a single parameter $q \geq 0$ and weights

$$w_q(y; x, p) = 2^q \mathbb{I}[d(y, p) \geq d(x, p)], \quad T_q(y \mid x, p) = \frac{\mathbb{I}[y \in N(x)] w_q(y; x, p)}{\sum_{z \in N(x)} w_q(z; x, p)}.$$

The three policies correspond to $q = 0$ (confused), $q = 1$ (afraid), and $q = 3$ (scared). Thus nonapproaching moves receive weights 1, 2, 8, respectively, and approaching moves receive weight 1. An isolated free cell has a self-transition of probability one. This models motion before capture; a known captured ghost is removed from the tracking distribution. For each active ghost the recursion is

$$\bar{b}_t(y) = \sum_x T_q(y \mid x, p_t) b_{t-1}(x), \quad b_t(y) = \frac{O(e_t \mid y, p_t) \bar{b}_t(y)}{\sum_z O(e_t \mid z, p_t) \bar{b}_t(z)}.$$

2 Implementation

- a. *Leave empty.*

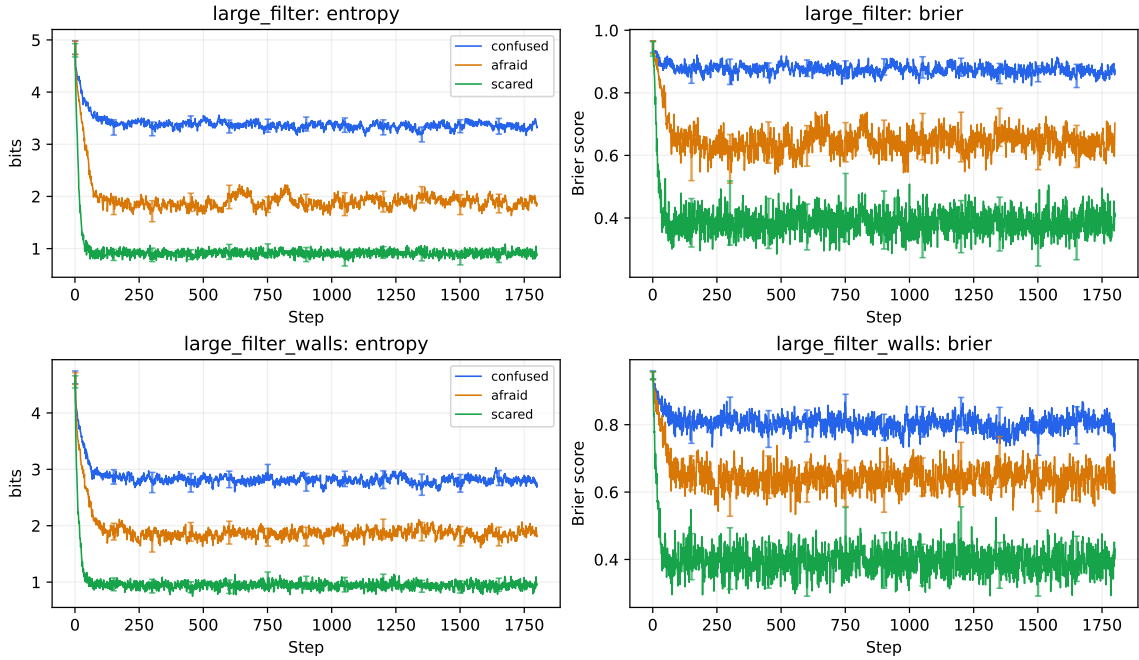
3 Experiment

- a. Uncertainty is Shannon entropy $H(b) = -\sum_{x: b(x) > 0} b(x) \log_2 b(x)$, measured in bits. Zero entropy means certainty; $\log_2 M$ is the maximum over M cells. Entropy measures confidence, not whether the confidence is correct. Active ghosts are averaged within a trial; captured ghosts are excluded.
- b. Quality is the Brier score $B(b, x^*) = \sum_x (b(x) - \mathbb{I}[x = x^*])^2$. It is zero for correct certainty and approaches two for wrong certainty. We also record $\sum_x b(x) d(x, x^*)$ and $b(x^*)$ for diagnostics. True positions are used only to compute evaluation metrics.
- c. We ran 30 trials per condition, with 3 independent ghosts per trial, on both supplied filter layouts. Each policy was tested at $v \in \{0.25, 1, 4\}$, using seed 193611. Runs used 1800 steps at $v = 0.25, 1$ and 9000 steps at $v = 4$; the longer runs measure the slower uncertainty decay at high noise. The default-variance

comparison is $v = 1$. To avoid early termination and survivor-selection bias, Pacman stays at its initial cell and capture is disabled in these controlled tracking experiments. Motion probabilities come from the supplied ghost policies evaluated through geometric observations without collision side effects; they are not sampled from the filter’s transition matrix. All ghosts move before each sensor update. The submitted filter and metric recorder are used unchanged. Separately, live-engine tests run the moving bonus controller with capture enabled on both layouts and every ghost policy (seeds 1 and 42, three ghosts).

Each plotted error bar is a pointwise 95% Student- t confidence interval across trial means, not across time steps or individual ghosts. The table averages the final 600 steps of each trial.

Layout	Policy	Entropy (bits)	Brier score
Open	confused	3.347 ± 0.033	0.870 ± 0.006
Open	afraid	1.902 ± 0.042	0.644 ± 0.009
Open	scared	0.910 ± 0.006	0.384 ± 0.003
Walls	confused	2.807 ± 0.085	0.800 ± 0.017
Walls	afraid	1.875 ± 0.070	0.645 ± 0.013
Walls	scared	0.943 ± 0.007	0.392 ± 0.003

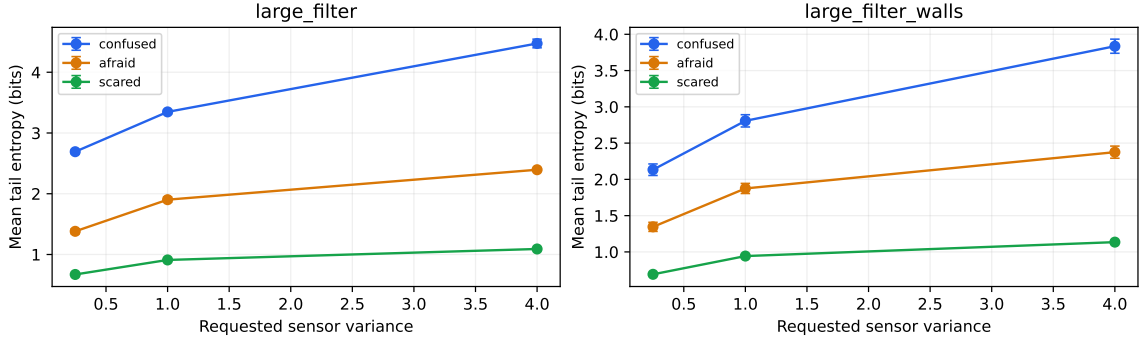


Convergence was checked by comparing the final two disjoint windows, paired by trial: 600 steps each at $v = 0.25, 1$, and 3000 steps each at $v = 4$. Across all 18 conditions, the largest absolute changes were 0.162 bits and 0.006 Brier units. At the required default variance, these maxima were only 0.014 bits and 0.004 Brier units. The high-noise confused policy still shows declining entropy; its variance comparison is a finite-duration result, not an equilibrium estimate. An additional confused-policy check used 27000 steps per trial at $v = 4$, with the same 30 trials, three ghosts and seed. Paired entropy changes (95% CI) were open: -0.101 ± 0.044 bits; walls: -0.047 ± 0.058 bits. Not all maps met the predeclared practical stability criterion $|\Delta| + \text{CI}_{95} \leq \epsilon$ for both metrics ($\epsilon_H = 0.05$ bits, $\epsilon_B = 0.01$). These additional results are stored separately in `results/high-noise-check.json`; the variance plot retains the original 9000-step study. The raw trial curves, individual confidence intervals and window changes are saved in `results/trials.npz` and `summary.json`. These are empirical stability diagnostics, not a proof of convergence for every map or a guarantee of exact localization.

- d. Increasing q makes moves away from Pacman relatively more likely. For example, with two approaching and two receding neighbors, the probability of a receding move is $1/2$, $2/3$ or $8/9$. With a fixed Pacman

near a corner, fearful ghosts concentrate near distant boundaries or local distance maxima. The measured default-noise entropy decreases from confused to afraid to scared in both layouts. Walls change available exits and create bottlenecks, so distance-only observations interact with map geometry. The table shows the resulting uncertainty and accuracy; the ordering does not imply that greater fear universally improves inference for arbitrary moving controllers.

- e. Increasing sensor variance broadens the measurement likelihood, retaining more plausible positions after each update. The experiment below shows greater tail entropy at higher variance for each policy on both maps. Transition information remains informative for fearful ghosts even when measurements are noisy. Parameter truncation matters: only $n/4$ is the actual variance, although the three values tested here are represented exactly.



- f. A controller can score legal candidate next cells using their capture probability and discounted proximity to belief mass. Our controller learns blocked neighbors from legal actions and remembers visit counts. Unknown cells remain traversable in its provisional map. For each legal move to z , it computes provisional maze distances and maximizes $\sum_g \sum_x b_g(x) e^{-\hat{d}(z,x)/4} + 2 \sum_g b_g(z) - 0.03 \text{ visits}(z)$. This favors likely captures and discourages repeated unproductive moves. The only state observations are its own position and legal actions; neither true ghost coordinates nor the wall-grid accessor is used. This is a heuristic controller, with no guarantee of optimal capture time.

- g. *Leave empty.*